Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 11: Approved Solutions

Sets, Subsets, Power Sets, and Cartesian Products

Covers Chartrand & Zhang, Sections 2.1 and 2.3

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Count the outermost braces first

Two habits that prevent most errors

1. **Before answering anything about a set, list its elements at the outermost level and count them.** For $\{1, \{2, 3\}\}$ the answer is two: the number 1 and the set $\{2, 3\}$.
2. **$\in$ takes an element on the left; $\subseteq$ takes a set on the left.** If unsure, ask whether the thing on the left appears *in* the list, or is *made of* things in the list.

Subset claims are quantified implications

$A \subseteq B$ means $\forall x, (x \in A \Rightarrow x \in B)$. Proving it takes an arbitrary element; disproving it takes one element of A outside B — the same asymmetry as Wednesday.

Three mistakes to expect

1. **$|\{\emptyset\}| = 0$.** It is 1. A set containing the empty set is not empty.
2. **Confusing $\in$ and $\subseteq$**, especially when elements are themselves sets.
3. **$A \times B = B \times A$.** The pairs are ordered; only the sizes match.

Part II: Practice Problems

Section A — Element or subset?

1. Let $A = \{0, \{0\}, \{0,1\}\}$. Determine whether each is true or false, with a one-line reason.

(a) $0 \in A$

(b) $\{0\} \in A$

(c) $\{0\} \subseteq A$

(d) $\{\{0\}\} \subseteq A$

(e) $1 \in A$

(f) $\emptyset \subseteq A$

Approved solution

A has **three** elements: 0, $\{0\}$, and $\{0,1\}$.
(a) True — 0 is listed.
(b) True — the set $\{0\}$ is itself listed as an element.
(c) True — its only element, 0, belongs to A.
(d) True — its only element, $\{0\}$, belongs to A.
(e) False — 1 appears *inside* the element $\{0,1\}$, but is not itself an element of A.
(f) True — the empty set is a subset of every set, vacuously.

Parts (b) and (c) together are the point. $\{0\}$ is both an element of A and a subset of A, for two different reasons. Both can hold at once, and neither implies the other.
Part (e) catches students who scan for a symbol rather than reading the structure.

Section B — Power sets

2. Write $\mathcal{P}(A)$ in full for each, and give $|\mathcal{P}(A)|$.

(a) $A = \{x, y, z\}$

(b) $A = \{\emptyset\}$

(c) $A = \{1, \{1\}\}$

Approved solution

(a) $\{\emptyset, \{x\}, \{y\}, \{z\}, \{x,y\}, \{x,z\}, \{y,z\}, \{x,y,z\}\}$, so $|\mathcal{P}(A)| = 8 = 2^3$.
(b) A has one element, so $\mathcal{P}(A) = \{\emptyset, \{\emptyset\}\}$ and $|\mathcal{P}(A)| = 2$.
(c) A has two elements, so $\mathcal{P}(A) = \{\emptyset, \{1\}, \{\{1\}\}, \{1, \{1\}\}\}$ and $|\mathcal{P}(A)| = 4$.

Part (b) is the one to check carefully. $\{\emptyset\}$ has one element, not zero, so its power set has two members. Compare $\mathcal{P}(\emptyset) = \{\emptyset\}$, which has exactly one.
Listing subsets by size — none, then one element, then two — makes it easy to see you have them all.

3. Answer each with a number and a one-line reason.

(a) If $|A| = 6$, what is $|\mathcal{P}(A)|$?

(b) If $|\mathcal{P}(A)| = 128$, what is $|A|$?

(c) How many elements does $\mathcal{P}(\mathcal{P}(\emptyset))$ have?

Approved solution

(a) $2^6 = 64$, by Theorem 2.17.
(b) $|A| = 7$, since $2^7 = 128$.
(c) Work outward. $\mathcal{P}(\emptyset) = \{\emptyset\}$, a set with one element. So $\mathcal{P}(\mathcal{P}(\emptyset)) = \mathcal{P}(\{\emptyset\})$ has $2^1 = 2$ elements:

$\emptyset$ and $\{\emptyset\}$.

Part (c) is done by evaluating the inner power set first and *counting* before going outward. Students who try to picture the whole thing at once usually stall.

Section C — Cartesian products

4. Let $A = \{1, 2\}$ and $B = \{a, b\}$.

(a) List $A \times B$ and $B \times A$.

(b) Are they equal? Are their cardinalities equal?

(c) List $A \times A$.

(d) If $|A| = 5$ and $|B| = 4$, give $|A \times B|$ and $|\mathcal{P}(A \times B)|$.

Approved solution

(a) $A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$ and $B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$.
(b) Not equal — $(1, a)$ is in the first and not the second, since pairs are ordered. **Cardinalities are equal**, both 4.
(c) $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$. Note $(1, 2)$ and $(2, 1)$ are distinct.
(d) $|A \times B| = 5 \cdot 4 = 20$, and $|\mathcal{P}(A \times B)| = 2^{20} = 1{,}048{,}576$.

Part (d) is worth a remark: a 5×4 table has twenty cells and over a million possible subsets of those cells. That growth is why counting arguments matter, and it is the same 2^n from Lecture 1 and from SAT last week.

Section D — Writing

5. In four to six complete sentences, explain the difference between $\{0\}$, $\emptyset$, and $\{\emptyset\}$. Give the cardinality of each and say why confusing them breaks power-set problems.

Approved solution

Model answer. $\emptyset$ is the empty set: it has no elements, so $|\emptyset| = 0$. The set $\{0\}$ contains exactly one element, the number zero, so $|\{0\}| = 1$. The set $\{\emptyset\}$ also contains exactly one element — but that element is the empty set rather than a number — so $|\{\emptyset\}| = 1$ as well, and it is a different set from $\{0\}$. The braces do real work: putting something inside braces creates a container holding that thing, and a container holding nothing is still a container. Anyone who treats $\{\emptyset\}$ as empty will compute its power set as $\{\emptyset\}$ with one element instead of the correct two, and errors of that kind compound fast once power sets are nested.

Full credit requires all three cardinalities and a clear statement that $\{\emptyset\}$ is not empty. The power-set consequence is the third point.

Section E — Challenge (optional)

6. Prove that $\emptyset \subseteq A$ for every set A, using the definition of subset as a quantified implication.

Approved solution

By definition, $\emptyset \subseteq A$ means

$$\forall x,\ \big(x \in \emptyset \ \Rightarrow\ x \in A\big).$$

Consider any x. The hypothesis $x \in \emptyset$ is false, because the empty set has no elements. An implication with a false hypothesis is true. So the implication holds for every x, and therefore $\emptyset \subseteq A$.

This is vacuous truth from Lecture 4, now doing real work. Notice that the argument never mentions A — which is why the result holds for *every* set, including $A = \emptyset$ itself.
Students often find this unsatisfying, as though something were smuggled in. Nothing is: the definition asks whether any element of $\emptyset$ escapes A, and there are no elements to escape.

Instructor notes

- **Problem 1(b) and 1(c)** deserve board time: $\{0\}$ is both an element and a subset of A, for different reasons. Students want it to be one or the other.
- **Problem 2(b)** catches $|\{\emptyset\}| = 0$. If widespread, fix it before Homework 5 — it corrupts every power-set answer downstream.
- **Problem 6** is where vacuous truth stops being a curiosity and becomes a tool. Pair it with Lecture 4's `all([])` example.
- **Problem 4(d)** lands 2^{20} on a 5×4 table. Good moment to recall the Day 1 password count and last week's SAT blow-up — same exponent, third context.
- Reserved for Homework 5: even-numbered exercises from Sections 2.1 and 2.3. Odd ones have printed answers and are self-check only.